22/09/2026 Implementation of SUBNETTING in CISCO PACKET TRACER simulator

Aim : To implement classless IP subnetting using CISCO PACKET TRACER & verify connectivity between diff. subnets.

Procedure :-

(i) Open CISCO Packet Tracer & create a new network topology.

(ii) Add routers, switches, & PCs. to the workspace.

(iii) Connect the devices using appropriate cables

(iv) Divide the network 192.168.1.0/24 into smaller subnets using the 127 subnet mask.

(v) Config. the IP addresses & subnet mask for the router interfaces.

(vi) Configure the PC's with suitable IP addresses, subnet masks & default gateway.

(vii) Configure the switch ports & router interface.

(viii) Use the Ping command to test connectivity b/w devices.

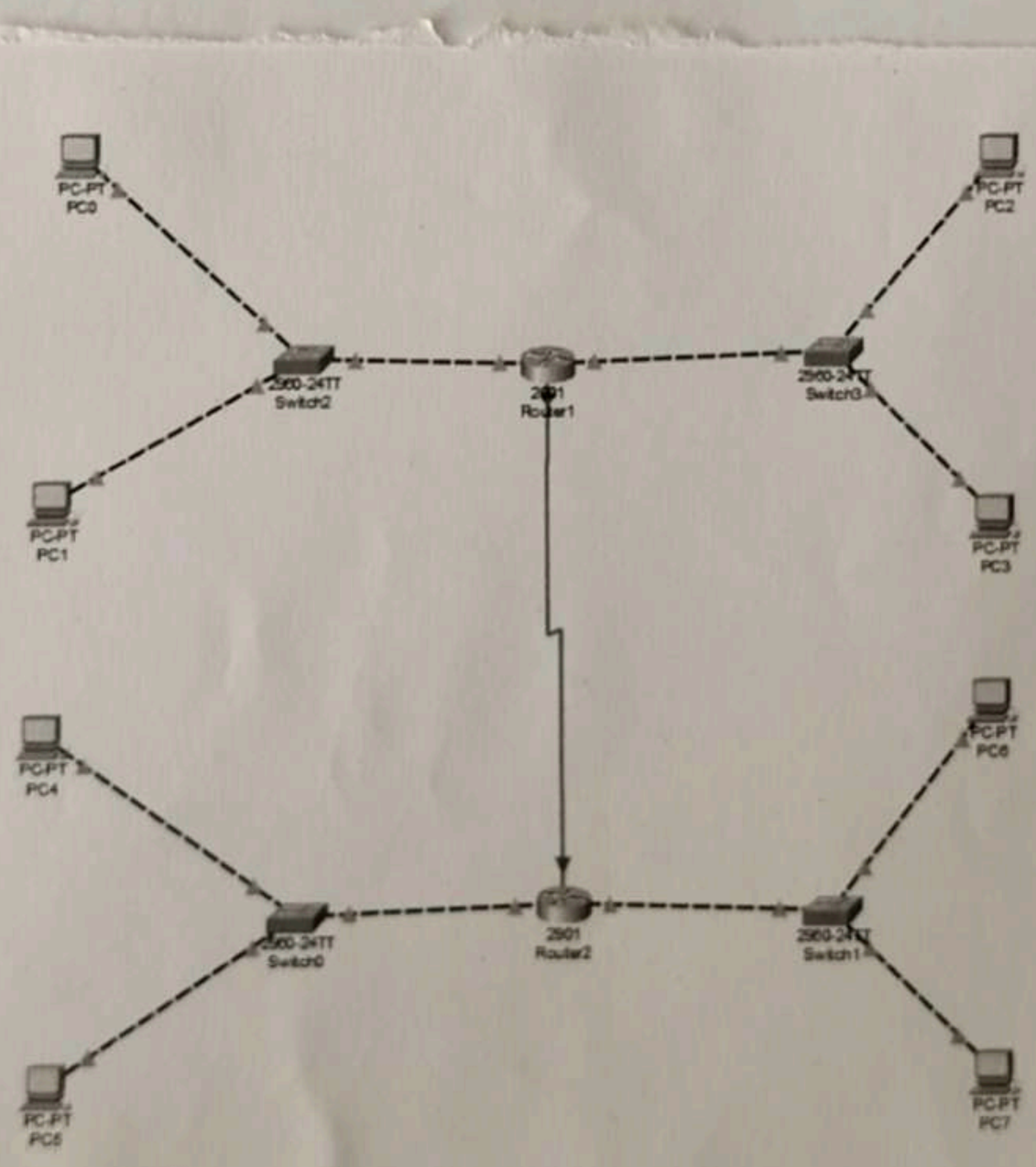
(ix) Verify the packets are successfully transmitted b/w the connected devices.

Advantages :-

* Efficient use of IP addresses.
* Reduces network congestion.
* Improves network management.

IP Addressing :-

Network	Subnet Mask	Router IP	PC IP-Range
192.168.1.0/27	255.255.255.224	192.168.1.1	192.168.1.11 -15
192.168.2.0/27	225.225.225.224	192.168.2.1	192.168.2.11 -15
192.168.3.0/27	225.225.225.224	192.168.3.1	192.168.3.11 -15
192.168.4.0/27	225.225.225.224	192.168.4.1	192.168.4.11 -15



Result : The classless IP subnetting was successfully implemented using CISCO PACKET TRACER.